Simulation-Based Inference

Introduction to Bayesian Inference

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Basics of Bayesian Inference

Likelihood

The likelihood is the probability density of the data y given the model parameters θ:

$$p(y \mid \theta)$$

Directly relates to common loss functions in machine learning:

$$L(\theta; y) = \log p(y \mid \theta)$$

Prior

The prior is the probability density of the model parameters θ *before* seeing the data y

$$p(\theta)$$

Prior information can come from several sources

Is the prior $p(\theta)$ really an *unconditional* distribution?

Directly relates to common regularization terms in machine learning:

$$R(\theta) = \log p(\theta)$$

Posterior

The posterior is the probability density of the model parameters θ *after* seeing the data y

$$p(\theta \mid y) = \frac{p(y \mid \theta)\, p(\theta)}{p(y)} = \frac{p(y, \theta)}{p(y)}$$

Directly relates to regularized loss functions in machine learning:

$$\begin{aligned} L_R(\theta; y) = \log p(\theta \mid y) &= \log p(y \mid \theta) + \log p(\theta) - \log p(y) \\ &= L(\theta; y) + R(\theta) - C \end{aligned}$$

Marginal likelihood

The marginal likelihood is the probability density of the data y after integrating out the model parameters θ:

$$p(y) = \int_{\Omega_\theta} p(y \mid \theta) \, p(\theta) \, d\theta$$

Bayes is hard because integration is hard

Ignorable in optimization:

$$\log p(y) = C$$

(Almost) every quantity of interest is an expectation over $p(\theta|y)$:

$$\mathbb{E}_{\theta|y}(f(\theta)) = \int f(\theta)\, p(\theta \mid y) \; \mathrm{d}\theta$$

Having obtained independent random samples $\{\theta^{(s)}\}$ from $p(\theta \mid y)$:

$$\frac{1}{S}\sum_{s=1}^{S} f(\theta^{(s)}) \sim \text{Normal}\left(\mathbb{E}_{\theta|y}(f(\theta)), \sqrt{\frac{\text{Var}_{\theta|y}(f(\theta))}{\text{S}}}\right)$$

How to deal with integrals?

How to deal with integrals?

Solve them...

- analytically: rarely possible
- numerically via quadrature: doesn't scale
- via brute force independent sampling: doesn't scale
- via smart dependent sampling (MCMC): great but still slow sometimes
- via parametric optimization (VI): fast but not so accurate

Example: Abundance of a genetic variant

- We are interested in measuring the abundance of a specific genetic variant.
- We want to model the rate θ (our parameter) that individuals of the population have the genetic variant of interest.

We need:

- The genetic data for each individual
- The likelihood / generative model (probability of the data given the parameters)
- The prior (probability of the parameters before seeing the data)

The binomial likelihood

- We call y_n the outcome of individual n
- The individual may have the genetic variant ($y_n = 1$) or not ($y_n = 0$)
- The genetic variant occurs with probability θ
- The genetic variant is absent with probability $1 - \theta$
- We have data on a total of N individuals

The Binomial likelihood for the number of individuals with the genetic variant $y := \sum_{n=1}^{N} y_n$:

$$p(y \mid \theta, N) = \binom{N}{y} \theta^y (1 - \theta)^{N-y}$$

Example: Marginal binomial likelihood with uniform prior

The marginal likelihood and thus the posterior are analytic in this case!

$$
\begin{aligned}
p(y) &= \int_0^1 \binom{N}{y} \, \theta^y \, (1-\theta)^{N-y} \times 1 \; d\,\theta = \binom{N}{y} \, B(y+1, N-y+1) \\
&= \binom{N}{y} \, \frac{\Gamma(y+1)\,\Gamma(N-y+1)}{\Gamma(N)} = \frac{N!}{y!(n-y)!} \, \frac{y!(N-y)!}{(N+1)!} \\
&= \frac{1}{N+1}
\end{aligned}
$$

Beta function:

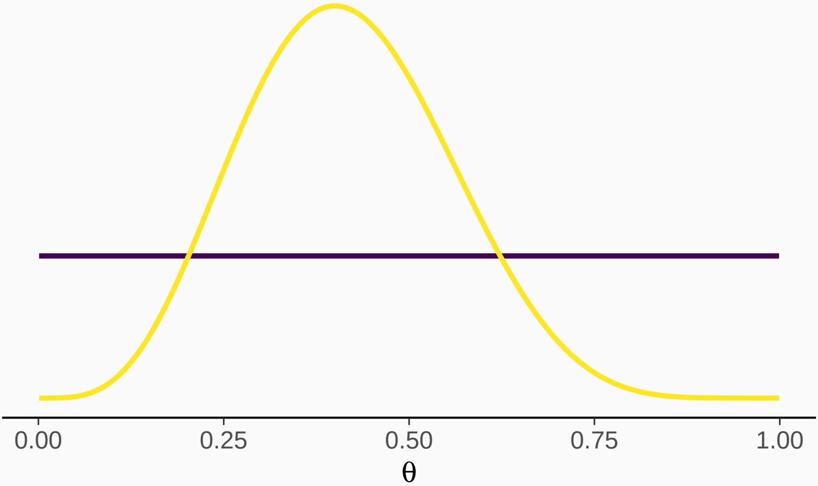
$$
B(a,b) := \int_0^1 \theta^{a-1}(1-\theta)^{b-1} \; d\,\theta
$$

Gamma function:

$$
\Gamma(z) := \int_0^\infty x^{z-1} \, \exp(-x) \, dx
$$

Posterior for a uniform prior

Suppose $y = 4$ out of a sample of $N = 10$ have been found to have the genetic variant of interest.



Beta-binomial models

Likelihood and prior:

$$y \mid \theta \sim \text{binomial}(N, \theta)$$
$$\theta \sim \text{beta}(\alpha, \beta)$$

Posterior:

$$\theta \mid y \sim \text{beta}(y + \alpha, N - y + \beta)$$

The beta prior is *conjugate* to the binomial likelihood

$\rightarrow$: Conjugacy implies getting an analytic solution

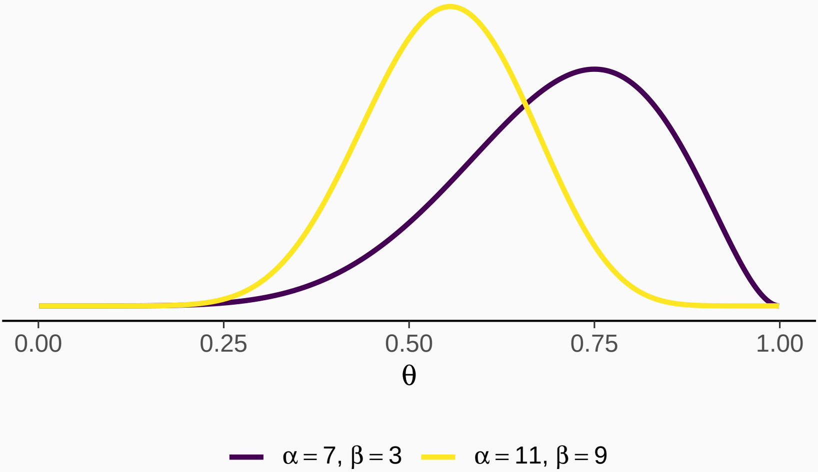
Prove conjugacy with only little math

Product of binomial likelihood and beta prior:

$$
\begin{aligned}
p(y, \theta) &= \binom{N}{y} \, \theta^{y} \, (1-\theta)^{N-y} \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1} \\
&= C \, \theta^{y+\alpha-1} \, (1-\theta)^{N-y+\beta-1} \\
&\propto \text{beta}(y+\alpha, N-y+\beta)
\end{aligned}
$$

Posterior for an informative prior

Again assume that $y = 4$ out of a sample of $N = 10$ have been found to have the genetic variant of interest.



Posterior summaries for an informative prior

- Data: $y = 4, N = 10$
- Prior: $\alpha = 7, \beta = 3$
- Posterior $\alpha' = 11, \beta' = 9$

Posterior mean:

$$\mathbb{E}(\theta) = \frac{\alpha'}{\alpha' + \beta'} = \frac{11}{11 + 9} = 0.55$$

Posterior standard deviation:

$$\mathrm{SD}(\theta) = \sqrt{\frac{\alpha' \beta'}{(\alpha' + \beta')^2 \, (\alpha' + \beta' + 1)}} \approx 0.11$$

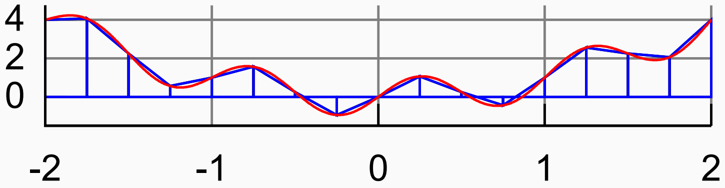
2.5% and 97.5% quantiles:

$$q_{0.025}(\theta) \approx 0.24 \qquad q_{0.975}(\theta) \approx 0.67$$

Trapezoidal rule for one-dimensions integration:

Interpolate the target function f by a series of straight lines passing through the points $(a, f(a))$ and $(b, f(b))$:

$$\int_a^b f(x)\, dx \approx (b-a)\left(\frac{f(a)+f(b)}{2}\right)$$



Numerical Integration via Quadrature

Multi-dimensional integration via recursive one-dimensional integration

First step with Trapezoidal rule:

$$\int_{a_2}^{b_2} \int_{a_1}^{b_1} f(x_1, x_2) \, dx_1 \, dx_2 \approx \int_{a_2}^{b_2} (b_1 - a_1) \left(\frac{f(a_1, x_2) + f(b_1, x_2)}{2} \right) \, dx_2$$

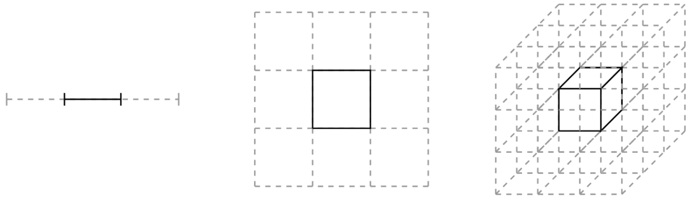
Second step with Trapezoidal rule:

- Do the same for x_2

Do you see any problems with this approach?

Why does numerical intergration not scale?

The number of support points for integral approximation scales exponentially with dimension



(a) (b) (c)

Common example for the curse of dimensionality

Monte Carlo estimation of the marginal likelihood

Idea: The marginal likelihood $p(y)$ is an integral so should be approximable via Monte Carlo estimation:

$$p(y) = \int p(y \mid \theta) p(\theta) d\theta = \mathbb{E}_\theta(p(y \mid \theta))$$
$$\approx \frac{1}{S} \sum_{s=1}^{S} p(y \mid \theta^{(s)}) \;\text{ for }\; \theta^{(s)} \sim p(\theta)$$

Question: Do you see potential problems with this approach?